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RESEARCH ARTICLE

The cold, hard truth: evidence from heterogeneous snowpack insulation effect on winter wheat yields in China

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Abstract

While previous scientific literature has found evidence that warming winters result in a loss of crop yields, recent studies suggest that temperature alone may not be the sole factor in determining yields. In this study, we investigate multiple climatic factors (freezing degree days (FDD) and snow cover fraction) to evaluate the heterogeneous impact of snowpack insulation on winter wheat yields in China. We find that a unit increase in FDD causes a loss of 2.37 kg/hectare in winter wheat yields. Furthermore, our results show that snowpack insulation has a statistically significant negative effect on winter wheat yields at lower and middle quantiles, but a statistically significant positive effect at higher quantiles of the winter wheat yields distribution, suggesting that snowpack insulation is important in maintaining higher winter wheat yields.

Keywords: heterogeneity analysis; snowpack insulation; warming winter; winter wheat yields

JEL classification: Q10; Q18; Q54

1. Introduction

Climate change research has shown that temperature warming prevails more in winter than in other seasons (see, e.g., Laing and Binyamin, 2013; Chen and Dong, 2021). Further, an increasing body of research emphasizes the heterogeneous impact and trends from regional and long-term winter warming stress (Yang *et al.*, 2007; Zou *et al.*, 2017; Qi *et al.*, 2022). This demands a strong interest in the evaluation of warming winter and its impact on the agricultural sector. However, recent research shows that it is important to consider rainfall and snow cover changes alongside temperature to evaluate the actual impact of warming winter on regional cropping patterns and yields (Zhu *et al.*, 2022).

As one of the largest countries in the world, China produces one-fourth of the world's grains and ranks first in the production of cereals.[1] According to the China

[1]See https://www.fao.org/china/fao-in-china/china-at-a-glance/en/.

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Agricultural Statistical Yearbook, the planting area of wheat is 23.38 million hectares, which is approximately 23.87 per cent of total cereal planting area, and wheat production is 134.25 million tons in 2020, which is approximately 21.77 per cent of the total cereal production. Among all kinds of wheat, winter wheat has the largest share, accounting for 96 per cent of total wheat production. Given the large planting area and production, a small change in winter wheat yields per hectare would lead to a substantial change in total crop yields.

As has been shown in previous scientific literature, agricultural production is sensitive to climate change (see, e.g., Ju *et al.*, 2013; Elliott *et al.*, 2014; Nelson *et al.*, 2014; Holzkämper, 2017). The average temperature in China has increased 1.2°C since 1960 and winter warming is four times the rate of summer warming (Piao *et al.*, 2010). Heavy precipitation would intensify with temperature increase in China at a rate of 6.25 per cent per degree of global warming (Zhang and Zhou, 2020). In addition to temperature and precipitation, researchers also found that a reduction in snowpack may lead to a reduction in soil insulation and existing vegetation in the face of harsh cold temperatures (Abys *et al.*, 2022; Li *et al.*, 2022; Zhu *et al.*, 2022). This could lead to soil exposure and early vegetation to low temperatures despite the recent warming winters. Moreover, diminishing snowpack may lower soil water content following spring, worsening crop yields.[2]

Though scholars have agreed on the fact that climate change affects agricultural output (see, e.g., Deschênes and Greenstone, 2007; Hertel and Rosch, 2010; Assunção and Chein, 2016; Kahsay and Hansen, 2016), most of the research conducted in this regard is based on heating degree days estimates from the Global North, whereas research from Global South countries is scarce due to the lack of granular crop yield data. Additionally, most research only considers the effect of heating degree days without taking into account that both heating and cooling degree days affect crop yield, and also underestimates the effect of precipitation and snow which is essential in the growth of crops. Utilizing satellite imagery of daily temperature, precipitation and snow depth, we are able to precisely identify winter wheat yields and corresponding climatic characteristics for Global South countries. In this paper, we address issues of temperature, precipitation and snow by combining measures of heating and cooling degree days, namely growing degree days (GDD) along with snow cover fraction (SCF) and freezing degree days (FDD). Furthermore, we provide evidence of snowpack insulation effect at different quantiles of winter wheat yields distribution using a novel method of quantile regression with fixed effects.

Specifically, we focus on the interaction of FDD and SCF across the three growing seasons (autumn, winter and spring) to estimate the effect of snowpack insulation on winter wheat yields using a panel data of 1,024 county-level administrative regions from 145 cities in 21 provinces in China, covering the period from 2001 to 2016. Based on a county fixed effects model, we include weather variables such as rainfall, temperature, and SCF to evaluate the impact of snowpack insulation on winter wheat yields while controlling for county-level fertilizers, soils and other county fixed effects to control for any time-invariant unobservables at the county level.

Our results show that a unit increase in FDD leads to a 2.37 kg/hectare winter wheat yield loss. Moreover, the increasing FDD causes consistent decreases in yields across

[2]For example, agricultural soil water availability in China relies heavily on snowmelt and diminishing snowpack further exacerbates droughts associated with changes in amount of snow water runoffs for crops in growing seasons.

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the winter wheat yields distribution. Thus, we illustrate that freezing temperature harms yields as seen previously in Zhu *et al.* (2022). Moreover, we find that snowpack insulation, estimated using the interaction of FDD*SCF, is estimated to have a statistically significant negative effect at lower and middle quantiles of the winter wheat yields distribution, and a statistically significant positive effect at higher quantiles of the distribution. Thus, it indicates that with an increasing SCF, on the one hand, the negative effect of FDD persists at lower and middle quantiles of the winter wheat yields distribution; on the other hand, the negative effect of FDD will be offset at higher quantiles of the distribution. Additionally, heterogeneous analysis demonstrates that the effect of snowpack insulation on winter wheat yields persists for plots with higher yields in regions with low economic development, low labor input, low irrigation areas, and climate of arid and warm temperature.

This paper makes important contributions to the existing literature. First, we provide one of the first nonlinear estimates of the effect of snowpack insulation on winter wheat yields in China using a quantile regression model with county fixed effects. Differently from Zhu *et al.* (2022) which uses county fixed effects models, we utilize the quantile regression to assess the effect of snowpack insulation at different quantiles of winter wheat yields distribution. By focusing on estimates from different quantiles, we are able to paint a more accurate picture of the snowpack insulation effect and provide insights for specific quantiles of winter wheat yields distribution. We show that snowpack insulation has a negative effect at lower and middle quantiles but a positive effect at higher quantiles of winter wheat yields distribution, though the literature has mainly demonstrated a positive snowpack insulation effect (Zhu *et al.*, 2022).

Second, we measure snowpack insulation as well as other climatic characteristics accurately and provide a unique opportunity to validate the snowpack insulation effect with high-quality satellite imagery. Using satellite imagery of daily temperature, precipitation, snow depth and annual fertilizer and soil textures, we are able to precisely identify counties' winter wheat yields and their environmental characteristics, especially for Global South countries without sufficient census datasets. This also sheds light on linking the scientific information with economic analysis, making it feasible to incorporate scientific information from satellite imagery with agricultural analysis. Moreover, though existing studies have investigated the heterogeneity effect of the snowpack insulation effect (Li *et al.*, 2022; Zhu *et al.*, 2022), we provide one of the first heterogeneity analyses from the perspective of economic development, labor input, irrigation and climate classification. Our findings suggest that temporary practices to increase SCF could be implemented, such as plastic mulch and artificial snow, which could be essential in keeping the snowpack insulation effect on winter wheat yields.

The rest of this paper is structured as follows. Section 2 presents the data. Section 3 presents the empirical strategy, and section 4 provides the main estimation results. Section 5 reports results from the robustness checks. Section 6 reports results from the heterogeneity analysis. Section 7 concludes by discussing important findings.

2. Data

2.1 Winter wheat yields

Our primary data consists of 1,024 county-level administrative regions from 145 cities in 21 provinces in China, covering the period from 2001 to 2016. We measure the county-level average winter wheat yields using the satellite imagery from Iizumi and Sakai (2020) based on the Chinese county-level administrative regions. The satellite imagery of crop

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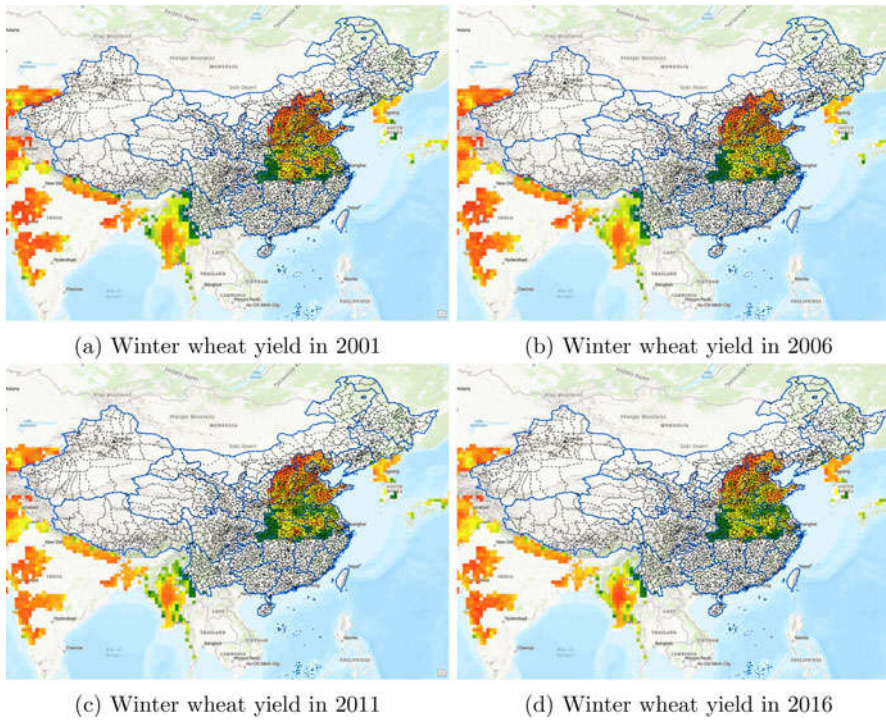


Figure 1. Winter wheat yields in China, (a) Winter wheat yield in 2001, (b) Winter wheat yield in 2006, (c) Winter wheat yield in 2011, (d) Winter wheat yield in 2016.
Notes: Solid blue lines represent boundaries of province-level administrative regions. Dashed black lines represent boundaries of county-level administrative regions. Red pixels represent high volume of winter wheat yields while green pixels represent low volume of winter wheat yields.

yields has been used in multiple analyses of yields (see, e.g., Zhang *et al.*, 2021; Sun *et al.*, 2022). Figure 1 presents the geographical locations of winter wheat growing regions in 2001, 2006, 2011 and 2016. As can been seen in figure 1, winter wheat is mostly cultivated in Northern China, such as in Shanxi Province, Shandong Province, Hebei Province and Henan Province, showing a good geographical coverage over a large territory in China.

Table 1 presents variable definitions and summary statistics. As shown in panel A, the mean and standard deviation of winter wheat yields[3] are 4,386.26 kg/hectare and 3,065.71 kg/hectare, respectively. This indicates a high yields variability in our sample, which is consistent with large territory coverage in figure 1. In addition to winter wheat yields, we match the county-level winter wheat yields with county-level climatic characteristics, fertilizers and soils. The rest of this section includes our data-construction methods for climatic variables as well as fertilizer and soil texture.

[3]We calculate the average winter wheat yields for all pixels of satellite imagery within county-level administrative regions. Additionally, we get the median winter wheat yields for all pixels of satellite imagery within county-level administrative regions for robustness checks.

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Table 1. Summary statistics

Variable	Definitions	Unit	N	Min	Median	Mean	Standard deviation	Max
Panel A: crop yields								
winterwheat	winter wheat yields	kg/hectare	1 5661	0.00	3816.29	4386.26	3065.71	16416.40
Panel B: climatic characteristics								
FDD	freezing degree days	Degrees Celsius	1 5661	0.00	80.89	225.97	372.88	3477.34
SCF	snow cover fraction	percentage	1 5661	0.00	0.08	0.18	0.25	1.00
autsnowfall	snowfall in autumn	mm	1 5661	0.00	0.05	4.63	9.85	120.46
autrainfall	rainfall in autumn	mm	1 5661	2.24	140.56	159.80	88.59	820.09
winsnowfall	snowfall in winter	mm	1 5661	0.00	14.93	19.84	22.72	641.68
winrainfall	rainfall in winter	mm	1 5661	0.00	18.78	49.97	69.33	434.02
sprsnowfall	snowfall in spring	mm	1 5661	0.00	0.22	4.48	17.58	982.47
sprrainfall	rainfall in spring	mm	1 5661	0.00	120.35	178.40	153.24	1261.40
autGDD_low	low growing degree days in autumn	Degrees Celsius	1 5661	31.94	823.13	780.84	129.07	910.00
autGDD_med	medium growing degree days in autumn	Degrees Celsius	1 5661	0.00	380.17	356.35	114.84	586.13

Continued.

Table 1. *Continued.*

Variable	Definitions	Unit	N	Min	Median	Mean	Standard deviation	Max
autGDD_high	high growing degree days in autumn	Degrees Celsius	15661	0.00	153.33	148.96	93.75	503.54
winGDD_low	low growing degree days in winter	Degrees Celsius	15661	0.00	158.82	169.50	124.40	424.08
winGDD_med	medium growing degree days in winter	Degrees Celsius	15661	0.00	21.82	46.76	56.94	275.46
winGDD_high	high growing degree days in winter	Degrees Celsius	15661	0.00	0.00	5.65	12.47	127.97
sprGDD_low	low growing degree days in spring	Degrees Celsius	15661	0.00	1271.70	1200.25	227.38	1520.04
sprGDD_med	medium growing degree days in spring	Degrees Celsius	15661	0.00	123.67	114.05	63.74	296.18
sprGDD_high	high growing degree days in spring	Degrees Celsius	15661	0.00	0.00	0.06	0.49	9.56
Panel C: fertilizer and soil texture								
nitrogenrate	Nitrogen fertilizer rate	kg/hectare	15661	0.00	214.15	193.11	80.21	349.27
clay100	proportion of clay particles (< 0.002 mm) in 0–100 cm layer	g/kg	15661	0.00	1074.20	1101.91	311.73	1965.18
sand100	proportion of sand particles (< 0.005 mm) in 0–100 cm layer	g/kg	15661	0.00	1326.94	1340.30	549.28	2863.84
soc100	soil organic carbon components in 0–100 cm layer	dg/kg	15661	0.00	435.69	508.62	305.02	2410.50

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2.2 Climatic characteristics

We use daily temperature to construct the FDD[4] to investigate the effect of freezing stress occurrence on winter wheat yields (Zhu *et al.*, 2022). Additionally, we also use daily temperature to construct the GDDs in spring (March to May), autumn (September to November) and winter (December to February) following equation (1) from Zhu *et al.* (2022) to control for the thermal effect during the winter wheat growing seasons. As is shown in panel B of table 1, the mean and standard deviation of FDD are 225.97°C and 372.88°C, respectively. For GDDs in autumn, winter and spring, we find that the mean and standard deviation for high GDD terms across three seasons are lower compared to low and medium GDD terms.[5]

To investigate the effect of snowpack insulation expected to reduce the freezing stress on winter wheat yields, we construct the SCF using FDD as well as daily snow depth (SND) as a ratio of days when both FDD>0 and SND> 2 cm, to days when FDD> 0, following equation (2) from Zhu *et al.* (2022).[6] In addition to the insulation effect, snow is one of the important water storage sources that provides soil moisture. In order to control for the effect of water supply from snowmelt, we partitioned precipitation into rainfall and snowfall to represent water supply from different sources on the basis of daily temperature and daily precipitation[7] following equation (3) from Zhu *et al.* (2022). Panel B in table 1 shows that the mean and standard deviation of SCF are 0.18 and 0.25, respectively. This indicates that on average 18 per cent of days with freezing stress occurrence (FDD> 0) have SND greater than 2 cm. We also find that average snowfall in winter is 19.84 mm, which is higher than counterparts in spring (4.48 mm) and autumn (4.63 mm). Differently from snowfall, average rainfall in spring (178.40 mm) and autumn (159.8 mm) is higher than that in winter (49.97 mm). The difference between rainfall and snowfall in winter is consistent with intuition that snow is heavier in winter, leading to the fact that a higher proportion of precipitation is from snowfall instead of rainfall in winter than in autumn and spring.

[4]We obtained daily temperature from the Copernicus Climate Change Service (ERA5: Fifth generation of ECMWF atmospheric reanalyses of the global climate. Copernicus Climate Change Service Data Store, available at https://cds.climate.copernicus.eu/). The daily FDD is calculated as follows:

$$FDD_d = \begin{cases} 0 - temp_d & \text{if} \quad temp_d < 0 \\ 0 & \text{if} \quad temp_d \geq 0 \end{cases}$$

where $temp_d$ is the daily temperature. FDD is calculated as the sum of daily degree days < 0°C as follows:

$$FDD = \sum_d FDD_d.$$

[5]While we include climatic variables, such as FDD, GDD, rainfall and snowfall as controls to capture their impact on crop yields, we also collect solar radiation, wind speed, and relative humidity sourced from Abatzoglou *et al.* (2018) using ERA5 Daily Land Aggregates data. Figure A11 in the online appendix shows estimates from our main regression including these variables. As is shown in figure A11, the consistent pattern across quantiles with those observed in figure 2 shows the robustness of our results.

[6]We obtained daily SND from the National Snow and Ice Center (Canadian Meteorological Centre) operational global daily snow depth analysis, available at https://nsidc.org/data/nsidc-0447/versions/1.

[7]We obtained daily precipitation from the Copernicus Climate Change Service (see footnote 4).

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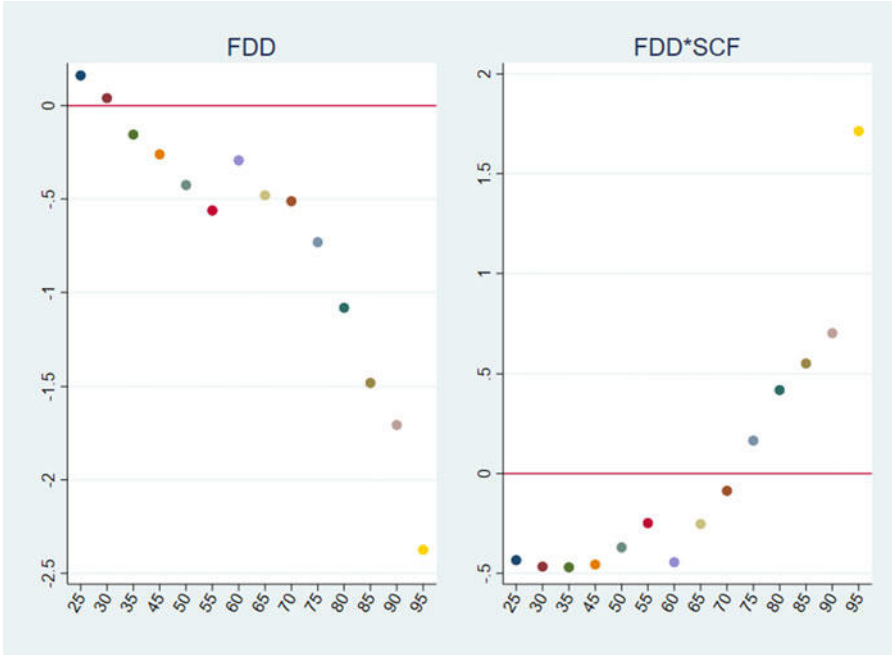


Figure 2. QRPD estimation results: FDD and FDD*SCF.
Notes: The dependent variable in all regressions is the mean of winter wheat yields. The left panel shows QRPD estimation results for FDD and the right panel shows QRPD estimation results for FDD*SCF.

2.3 Fertilizer and soil

As has been shown in previous literature, winter wheat yields are significantly affected by nitrogen fertilizer rates (Litke *et al.*, 2018; Tabak *et al.*, 2020) and soil textures (Thaler *et al.*, 2012; Johnen *et al.*, 2014). We include the county-level nitrogen fertilizer rates[8] and proportions of soil textures,[9] such as clay particles, sand particles and soil organic carbon components, to control for the effect of nitrogen fertilizer use and soil components on winter wheat yields (Zhu *et al.*, 2022). As shown in panel C of table 1, the mean and standard deviation of the nitrogen fertilizer rate are 193.11 kg/hectare and 80.21 kg/hectare, respectively. The mean and standard deviation of clay particles, sand particles, and organic carbon components are 1,101.91 g/kg and 311.73 g/kg, 1,326.94 g/kg and 549.28 g/kg, and 508.62 dg/kg and 305.02 dg/kg, respectively.[10]

[8]Following Mueller *et al.* (2012), we obtained nitrogen fertilizer from EarthStat, available at http://www.earthstat.org/nutrient-application-major-crops/.

[9]Following Poggio *et al.* (2021), we obtained the soil textures from the International Soil Reference and Information Centre, available at https://www.soilgrids.org/.

[10]We exclude time invariant variables, such as nitrogen fertilizer rates, clay particles, sand particles and soil organic carbon components from the main regression for robustness check. As shown in figure A8 in the online appendix, our estimates remain consistent with those in figure 2, demonstrating robustness of our findings.

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3. Empirical strategy

3.1 Conceptual framework

Our primary empirical strategy relies on quantile regression for panel data (QRPD), which is a special case of the generalized quantile estimator in which the estimator addresses a foundational problem of including fixed effects (Powell, 2014a, 2020, 2022). Our choice of an empirical framework stems from the nature of the research objective and variables of interest. Our primary objective is to explore the heterogeneous impact of snowpack insulation on winter wheat yields. This indicates a potentially varying effect at different points in the conditional distribution of the outcome variable. Furthermore, the snowpack exhibits significant inter- and intra-annual variability (Mankin and Diffenbaugh, 2015). Therefore, the variability is a crucial aspect of evaluating the impact of snowpack insulation on crop yields (Huning and AghaKouchak, 2020; Qin *et al.*, 2020).

Previous literature has shown that heterogeneous effects are useful information, but usually missed by mean regression techniques (Bitler *et al.*, 2006; Powell, 2014a; Powell and Wagner, 2014). Moreover, in recent advancements, there has been an impetus to combine the advantages of panel data regression as a tool to account for the unobserved heterogeneity with the ability of quantile regression to estimate robust estimates accounting for outliers (Koenker, 2004; Canay, 2011; Galvao Jr, 2011; Ponomareva, 2011; Kato *et al.*, 2012). However, combining the quantile and panel regression, these methods exploit additive fixed effects in the quantile function at the cost of loss of generality in the quantile regression inference and interpretation. In a novel approach, Powell (2022) extends this approach to build non-additive fixed effects with conventional quantile regression advantages such as a nonseparable disturbance term.Powell (2022) provides an estimator in an instrumental variable framework for fixed effects that are consistent with the small time period and are feasible to estimate in standard econometric software.[11] Powell (2022) presented an estimator which utilizes fixed effects for identification purposes, while the results are interpreted as traditional cross-sectional quantile regression estimates. Fixed effects enable arbitrary correlation with the level of variables of interest, accounting for unobserved county-level differences in non-parametric specification.

In summary, our proposed empirical strategy has two key advantages. Firstly, our dataset shows that the winter wheat yields exhibit positive skewness, which indicates a potentially nonlinear relationship.[12] Previous researchers have argued that climatic variables such as temperature and rainfall have nonlinear relationships with crop yields (Schlenker and Roberts, 2009; Burke *et al.*, 2015; Proctor, 2021). Therefore, we believe that the investigation of the winter wheat yields should take explicitly into account the distribution of outcome variables. However, several applied economic research studies regarding the effect of snowpack insulation on winter wheat yields do not explicitly model nonlinear relationships. We suggest that the quantile relationship is better suited for modeling climatic variables such as snowpack insulation and winter wheat yields. Furthermore, our proposed approach is robust to accounting for outliers in the data (Cameron and Trivedi, 2022; Powell, 2022).

Secondly, we estimate the heterogeneous relationship between winter wheat yields and snowpack insulation with an explicit quantile function. The approach is semi-parametric in terms of not having parametric distribution of regression errors and its

[11] In this paper, we utilize the Stata package called QRPD. Please refer to Baker (2016) and Powell (2014b).
[12] Figure A1 in the online appendix shows the distribution of winter wheat yields.

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departure from the normality assumptions (Cameron and Trivedi, 2022). Moreover, the nonlinear relationship between winter wheat yields and snowpack insulation has varying effects throughout the outcome distribution, and previous authors have found that snowpack insulation heterogeneously impacts crop yields (Li *et al.*, 2022; Zhu *et al.*, 2022). To our knowledge, we are one of the first to estimate the effect of snowpack insulation using a distribution of crop yields.

3.2 Model specification

We begin with an estimation of the two-way fixed effects (TWFE) model to estimate the relationship between the winter wheat yields and snowpack insulation effect as well as other climatic characteristics alongside the agro-economic variables such as fertilizer, and soil attributes. Our main estimating equation is specified as follows:

$$\begin{aligned}\text{WheatYields}_{i,t} = {} & \alpha_1 \text{FDD}_{i,t} + \alpha_2 \text{FDD}_{i,t} \times \text{SCF}_{i,t} + \sum_{j=1}^{3} \alpha_{3,j} \text{Rainfall}_{i,t}^{j} \\ & + \sum_{j=1}^{3} \alpha_{4,j} \text{Snowfall}_{i,t}^{j} + \sum_{j=1}^{3} \alpha_{5,j} \text{GDD}_{i,t}^{j} + \alpha_{11} \text{Fert}_i + \alpha_{12} \text{Sand}_i \\ & + \alpha_{13} \text{Clay}_i + \alpha_{14} \text{OM}_i + \text{procitydis}_i + \text{time}_t + e_{i,t},\end{aligned} \tag{1}$$

where WheatYields is the winter wheat yields. The $\text{FDD}_{i,t}$ is the FDD in county i at year t. The $\text{FDD}_{i,t} \times \text{SCF}_{i,t}$ is the interaction term between FDD and SCF in county i at year t. The $\text{Snowfall}_{i,t}^{j}$, $\text{Rainfall}_{i,t}^{j}$ and $\text{GDD}_{i,t}^{j}$ are the snowfall, rainfall and GDD in county i at year t for season j, respectively. The Fert_i, Sand_i, Clay_i and OM_i are the nitrogen fertilizer rates, clay particles, sand particles and soil organic carbon components in county i, respectively. The procitydis_i is county FEs, controlling for any time-invariant unobserved heterogeneity at the county level. The time_t denotes the year FEs.[13] We primarily focus on the estimates of α_1 and α_2 which represent the relationship between winter wheat yield with FDD and FDD*SCF.

In TWFE, the estimates reflect the average relationship, which is further refined in quantile regression as introduced by Koenker and Bassett Jr (1978). The following equation illustrates how the impact of the variable of interest changes based on total distribution:

$$y_i = \mathbf{d}_i' \beta \left(u_i^*\right).$$

To build on the framework from Powell and Wagner (2014), the structural quantile function (SQF) can be written as

$$S_{wheatyield_i} = \mathbf{d}_i' \beta(\tau), \quad \tau \in (0, 1).$$

The SQF provides the τ th quantile of $y_{\mathbf{d}} = \mathbf{d}_i' \beta(u^*)$ for a fixed $\mathbf{d}$ and a randomly-selected $u^* \sim U(0,\ 1)$. In our context, the SQFs provide the distributions of the outcome variable

[13] We use province-year and province-year quadratic fixed effects in the main regression for robustness check. As shown in figures A9 and A10 in the online appendix, our estimates are consistent with those presented in figure 2, showing the robustness of our results.

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winter wheat yield. More specifically, we aim to account for covariates and fixed effects as shown in equation (1) as follows:

$$\begin{aligned} S_{wheatyield_{it}} = {} & \alpha_1 \mathrm{FDD}_{i,t} + \alpha_2 \mathrm{FDD}_{i,t} \times \mathrm{SCF}_{i,t} + \sum_{j=1}^{3} \alpha_{3,j} \mathrm{Rainfall}_{i,t}^{j} \\ & + \sum_{j=1}^{3} \alpha_{4,j} \mathrm{Snowfall}_{i,t}^{j} + \sum_{j=1}^{3} \alpha_{5,j} \mathrm{GDD}_{i,t}^{j} + \alpha_{11} \mathrm{Fert}_i + \alpha_{12} \mathrm{Sand}_i \\ & + \alpha_{13} \mathrm{Clay}_i + \alpha_{14} \mathrm{OM}_i + \mathrm{procitydis}_i + \mathrm{time}_t + e_{i,t}, \quad \tau \in (0,1). \end{aligned} \tag{2}$$

We present equation (2) building on Powell (2014a) with the aim of estimating the α_1 and α_2, which represents the relationship between the winter wheat yield with FDD and FDD*SCF.

4. Results

4.1 The impact of FDD

To paint an accurate picture for different quantiles, we choose 15 quantiles from the 25th to the 95th (i.e., 25th, 30th, 35th, 40th, 45th, 50th, 55th, 60th, 65th, 70th, 75th, 80th, 85th, 90th, 95th) following equation (2). Figure 2 provides a graphical representation of the QRPD estimation results for FDD and FDD*SCF in the left and right panels, respectively.[14] The left panel shows that FDD is estimated to have a statistically significant negative effect throughout the entire distribution of winter wheat yields with a few exceptions at lower quantiles (i.e., the 25th and 30th quantiles). This result indicates that a higher occurrence of freezing stress leads to a greater loss of winter wheat yields, which is also consistent with the findings from Zhu *et al.* (2022). This result is also aligned with previous literature that freezing temperature has a negative impact on crop yields (Verón *et al.*, 2015; Zhang *et al.*, 2021). Additionally, we find that the magnitudes of the FDD estimates are increasing from lower to higher quantiles of the winter wheat yields distribution, suggesting that higher occurrence of freezing stress decreases winter wheat yields in counties with higher winter wheat yields.[15] Based on the FDD estimate at the 95th quantile in column (15) in table A1 (online appendix), we find that a unit increase in FDD leads to a loss of 2.37 kg/hectare in winter wheat yields.[16]

[14]Table A1 in the online appendix provides numeric QRPD estimation results for FDD and FDD*SCF.

[15]Table A3 and figure A3 in the online appendix present a numeric and graphical representation of the QRPD estimates for FDD and FDD*SCF using the median of winter wheat yields as a robustness check and no substantial changes are observed.

[16]Using standardized FDD and SCF, table A2 in the online appendix shows that a one-standard deviation increase in FDD reduces winter wheat yields by 17.37 per cent of a standard deviation for the 95th quantile. Figure A2 in the online appendix is the corresponding figure across all quantiles. Given the large magnitude of FDD standard deviation (372.88◦C), the standardized estimation results show that a small increase in the FDD variation would lead to a substantial decrease in winter wheat yields. Furthermore, our estimates show that a one-unit increase in FDD leads to a 2.37 kg/ha loss in yields, which is equivalent to 0.05 per cent of mean winter wheat yields. To benchmark the magnitude of our estimated effects in the literature, Zhu *et al.* (2022) estimate that a unit increase in FDD is associated with a 0.051 ± 0.0060 per cent yield decline. Our estimated effects are consistent with previous findings.

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4.2 The impact of SCF

As indicated in equation (2), we investigate the snowpack insulation effect on winter wheat yields by accounting for the interaction term between FDD and SCF. As shown in the right panel of figure 2, we find that FDD*SCF is estimated to have a statistically significant negative effect at lower and middle quantiles of the winter wheat yields distribution (from the 25th to 70th quantiles), and a statistically significant positive effect at higher quantiles of the distribution (from the 75th to 95th quantiles), indicating that with an increasing SCF, on the one hand, the negative effect of FDD persists in lower and middle quantiles of the winter wheat yields distribution, while on the other hand, the negative effect of FDD will be offset at higher quantiles of the distribution. This result shows that the snowpack insulation effect could reduce the impact of freezing temperature, measured by FDD, at high quantiles of the winter wheat yields distribution; it is also different from Zhu *et al.* (2022) where the interaction term of FDD*SCF is estimated to have a statistically significant positive impact on wheat yields. Based on the estimate of FDD*SCF at the 95th quantile in column (15) in table A1 (online appendix) and the mean of FDD in table 1, we find that a one-percent increase in SCF leads to an increase of 3.87 kg/hectare in winter wheat yields.

4.3 The impact of GDD and snowfall

Figure 3 shows the results of the estimation of QRPD for all GDD terms.[17] Low GDD in autumn, medium GDD in autumn, low GDD in winter and high GDD in spring are estimated to have a predominantly statistically significant positive effect on winter wheat yields throughout the entire distribution of winter wheat yields. Based on estimates for high GDD in spring, we find that a unit increase in high GDD in spring increases winter wheat yields by 110 kg/hectare at the 50th quantile. This result is different from the findings from Zhu *et al.* (2022) where a high GDD in the spring is associated with a loss of wheat yields.

The effect of water supply from snowmelt is represented by snowfall terms in autumn, winter and spring. As shown in the top three panels in figure 4,[18] we find that snowfall in autumn and winter is estimated to have a statistically significant positive effect on wheat wheat yields throughout the entire distribution. The magnitudes of snowfall estimates in autumn and winter are increasing from lower quantiles to higher quantiles. This result suggests a supportive role of snowfall on wheat growth by increasing moisture during the cold seasons, which is consistent with the findings of Zhu *et al.* (2022).

5. Robustness checks

5.1 Alternative snow cover fraction

In the previous section, we constructed SCF as a ratio of days when both FDD$>$ 0 and SND$>$ 2 cm. To investigate whether threshold selection is a concern, we report estimates from alternative thresholds of SND, such as 5 cm, 10 cm, 15 cm and 20 cm. Figure A6 (online appendix) shows estimates for FDD and FDD*SCF with thresholds of SND for 5

[17]Figure A4 in the online appendix presents a graphical representation of QRPD estimates for all GDD terms using the median of winter wheat yields as a robustness check and no substantial changes are observed.

[18]Figure A4 in the online appendix presents a graphical representation of QRPD estimates for snowfall and rainfall terms using the median of winter wheat yields as a robustness check and no substantial changes are observed.

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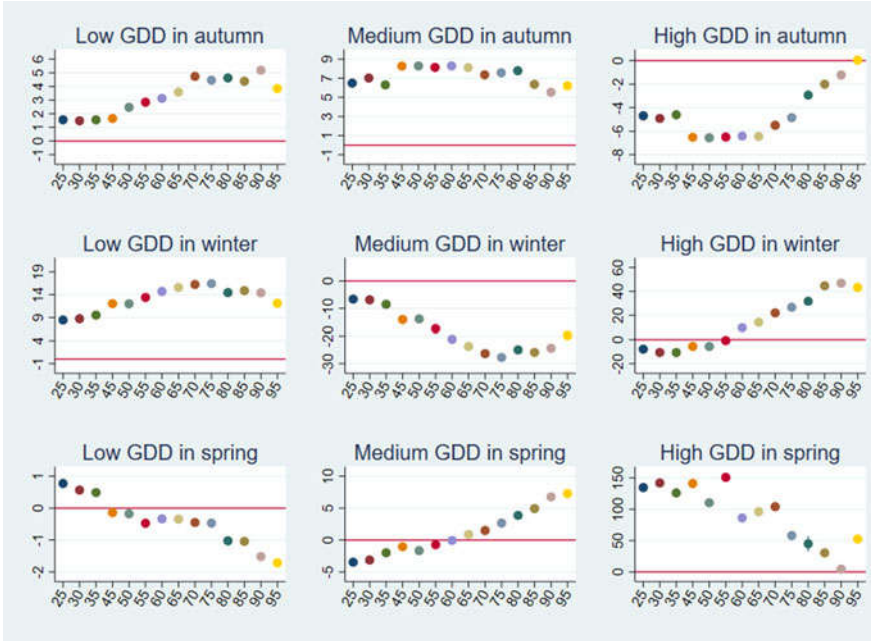


Figure 3. QRPD estimation results: GDDs.
Notes: The dependent variable in all regressions is the mean of winter wheat yields. The top three panels show QRPD estimation results for GDD in autumn, the middle three panels show QRPD estimation results for GDD in winter and the bottom three panels show QRPD estimation results for GDD in spring.

cm, 10 cm, 15 cm and 20 cm. As can be seen in figure A6, the estimates of FDD in panels (a), (b), (c) and (d) are significantly negative, which is consistent with the results from figure 2. As SND thresholds are higher, the estimates for FDD*SCF are predominantly significantly positive. Though the estimates for FDD*SCF in figure A6 are different from the results in figure 2, the snowpack insulation effect still persists to offset the negative effect of FDD. The difference is due to a higher threshold of SND, which makes the effect of snowpack insulation stronger.

5.2 Alternative growing degree days

In the previous section, we constructed season-specific GDDs following thresholds used in Zhu *et al.* (2022). To alleviate the concern about threshold selection, we construct new season-specific GDDs following thresholds used in Tack *et al.* (2015). The thresholds to categorize low, medium and high GDDs are 0° C/18° C/23°C for autumn, 0° C/5° C/10°C for winter, and 0° C/17° C/28°C for spring. Furthermore, we also construct new non-season-specific GDDs following the thresholds of 0°C/26°C/35°C used in Zhu *et al.* (2022). In figure A7 in the online appendix, panel (a) shows estimates for FDD and FDD*SCF using new season-specific GDDs and panel (b) shows estimates for FDD and FDD*SCF using new non-season-specific GDDs. Results from figure A7 show that FDD is estimated to have a statistically significant negative effect, which is consistent with findings from figure 2. From panel (a) in figure A7, FDD*SCF is estimated to have a

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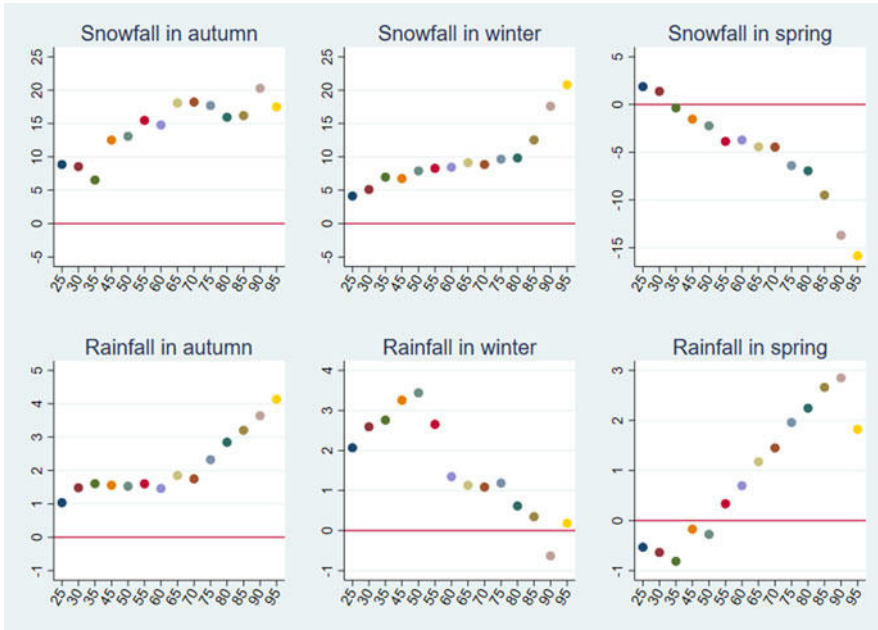


Figure 4. QRPD estimation results: snowfall and rainfall.
Notes: The dependent variable in all regressions is the mean of winter wheat yields. The top three panels show QRPD estimation results for GDD in autumn, the middle three panels show QRPD estimation results for GDD in winter and the bottom three panels show QRPD estimation results for GDD in spring.

statistically significant positive effect at higher quantiles of the winter wheat yields distribution (from the 75th to 95th quantiles), which is aligned with the results from figure 2. From panel (b) in figure A7, FDD*SCF is estimated to have a predominantly statistically significant positive effect, showing the existence of the snowpack insulation effect.

6. Heterogeneity analysis

6.1 Economic development

Previous literature has shown that nightlight is a good proxy for economic development (see, e.g., Chen and Nordhaus, 2011; Nordhaus and Chen, 2015; Gibson *et al.*, 2021). We construct the annual average of nightlight intensity at county-level administrative regions from 2001 to 2016[19] and categorize regions as low and high nightlight regions based on the median of nightlight intensity. Figure 5 shows QRPD estimation results for FDD and FDD*SCF for low and high nightlight regions. Panel (a) shows that FDD is estimated to have a statistically significant negative effect on winter wheat yields for regions with low nightlight intensity but a positive effect for regions with high nightlight intensity throughout the entire distribution of winter wheat yields. Panel (b) shows that FDD*SCF is estimated to have a statistically significant negative effect at lower and middle quantiles of the winter wheat yields distribution (from the 25th to 65th quantiles), and a positive effect at higher quantiles of the distribution (from the 70th to 95th

[19]We obtained annual nightlight intensity from Li *et al.* (2020).

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quantiles). Comparing the results in figure 5 to the results in figure 2, we find that the snowpack insulation effect persists at higher quantiles of winter wheat yields distribution for regions with low nightlight intensity (FDD< 0 and FDD*SCF> 0), indicating that a heterogeneous effect of snowpack insulation exists. One of the possible explanations is that agricultural activities are less likely to be invested in regions with low economic development (i.e., low nightlight intensity), preserving the original natural effect of snowpack insulation.

6.2 Labor input

As is shown in Tongwei *et al.* (2020) and Meiyan (2010), labor input is one of the important factors in agricultural productivity. To investigate the heterogeneous effect of snowpack insulation based on labor input, we construct the annual average population density at the county-level administrative regions from 2001 to 2016[20] and categorize them as low and high population density regions based on the median of population density. Figure 6 shows QRPD estimation results for FDD and FDD*SCF for low and high population density regions. Panel (a) shows that FDD is estimated to have a statistically significant negative effect on winter wheat yields for low population density regions but a statistically significant positive effect for high population density regions throughout the entire distribution of winter wheat yields. Panel (b) shows that FDD*SCF is estimated to have a significantly negative effect on winter wheat yields throughout the entire distribution with exceptions at the 90th and 95th quantiles for low population density regions. Comparing FDD and FDD*SCF estimates in figure 6 to those in figure 2, we find that the snowpack insulation effect persists at higher quantiles of the winter wheat yields distribution in low population density regions, indicating a heterogeneous effect of snowpack insulation exists. One of the possible explanations is that farmers are less likely to increase labor input in winter wheat cultivation in low population density regions, keeping the original natural effect of snowpack insulation.

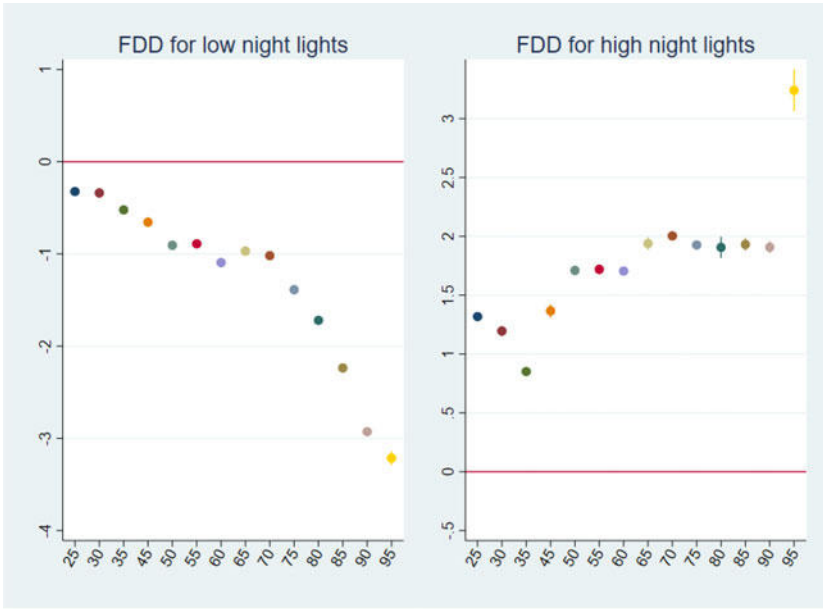
6.3 Irrigation intensity

As shown in previous scientific literature (Sun *et al.*, 2006; Liu *et al.*, 2007; Zeng *et al.*, 2021), irrigation plays an important role in winter wheat yields. We construct the annual average irrigation area at the county-level administrative regions from 2001 to 2016[21] and categorize regions as low and high irrigation regions based on the median of irrigation area. Figure 7 presents QRPD estimation results for FDD and FDD*SCF for low and high irrigation regions. Panel (a) shows that FDD is estimated to have a statistically significant negative effect on winter wheat yields for low irrigation regions but a statistically significant positive effect for high irrigation regions throughout the entire distribution of winter wheat yields. Panel (b) shows that FDD*SCF is estimated to have a significantly positive effect on winter wheat yields at higher quantiles of winter wheat yields distribution for both low (90th and 95th quantiles) and high (80th to 95th quantiles) irrigation regions. Based on FDD and FDD*SCF estimates in low irrigation regions which are consistent with estimates of FDD and FDD*SCF from figure 2, we find that the snowpack insulation effect persists at higher quantiles of the distribution for the low

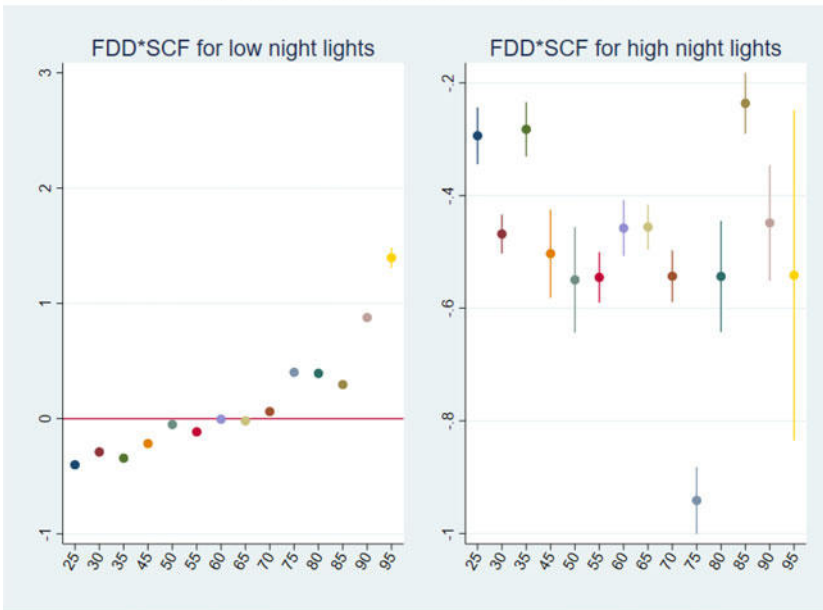
[20] We obtained annual population from Oak Ridge National Laboratory, available at https://landscan.ornl.gov/.

[21] We obtained annual irrigation area from Zhang *et al.*, 2022.

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(a) QRPD estimation results: FDD



(b) QRPD estimation results: FDD*SCF

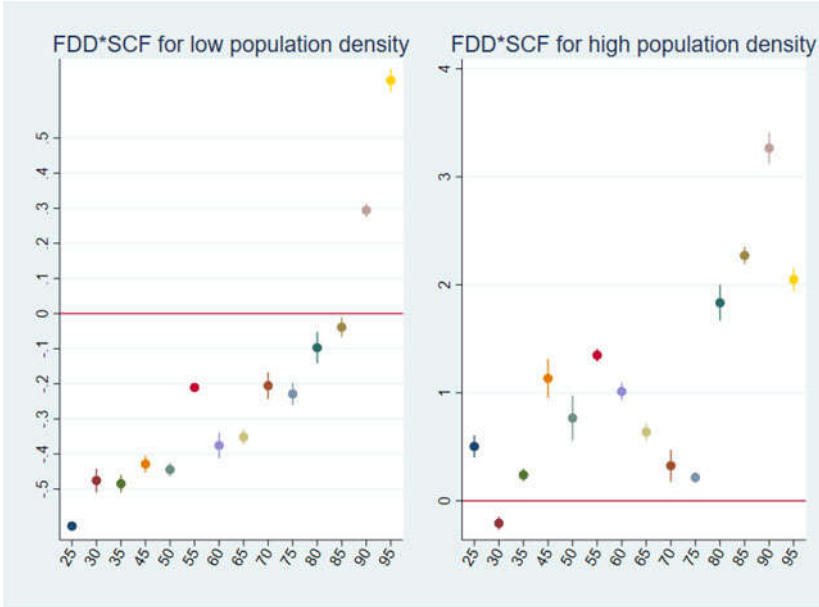
Figure 5. Heterogeneity analysis: nightlight, (a) QRPD estimation results: FDD, (b) QRPD estimation results: FDD*SCF.

Notes: The dependent variable in all regressions is the mean of winter wheat yields. The top two panels show QRPD estimation results of FDD for low and high nightlight intensity. The bottom two panels show QRPD estimation results of FDD*SCF for low and high nightlight intensity.

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(a) QRPD estimation results: FDD



(b) QRPD estimation results: FDD*SCF

Figure 6. Heterogeneity analysis: population density.
Notes: The dependent variable in all regressions is the mean of winter wheat yields. The top two panels show QRPD estimation results of FDD for low and high population density. The bottom two panels show QRPD estimation results of FDD*SCF for low and high population density.

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irrigation regions, suggesting a heterogeneous effect of snowpack insulation exists. One of the possible explanations is that high irrigation regions are more likely to be affected by artificial irrigation, distorting the original natural effect of snowpack insulation.

6.4 Climate classifications

As shown in figure 1, winter wheat is cultivated in a large territory in China, covering different climatic regions from warm to snow as well as humid to arid. To investigate the heterogeneous effect of climate classifications, we split regions into three categories: arid, temperature and perhumid.[22] Figure 8 shows QRPD estimation results for FDD and FDD*SCF for the climate of arid, temperature and perhumid. As can be seen in panel (a), FDD is estimated to have a statistically significantly negative effect on winter wheat yields throughout the entire distribution for all three climate classifications. Panel (b) shows that FDD*SCF is estimated to have a predominantly statistically significant positive effect on winter wheat yields for the climate of arid and temperature but a predominantly negative effect for perhumid. Based on FDD and FDD*SCF estimates in figure 8, we find that the snowpack insulation effect persists for the climate of arid and temperature, indicating a heterogeneous effect of snowpack insulation for different climate classifications.

7. Discussion and conclusion

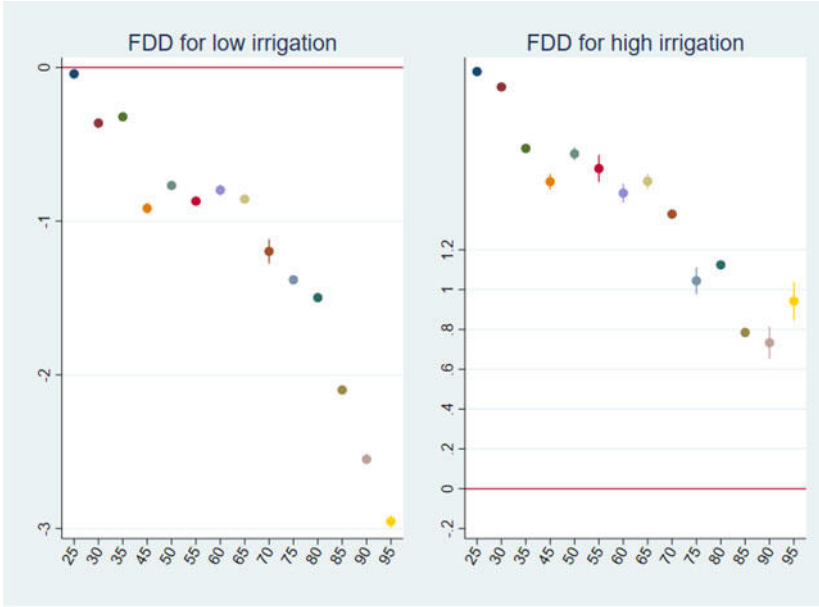
In this paper, we investigate the heterogeneous effect of snowpack insulation on winter wheat yields in China. Taking advantage of satellite imagery, we construct a county-level panel data, including winter wheat yields, climatic characteristics, fertilizer and soil textures to investigate the effect of snowpack insulation on winter wheat yields. Using the QRPD with additive fixed effects such as year and county fixed effects, we reveal the varying effects of snowpack insulation at different quantiles of the winter wheat yields distribution. In addition, we account for economic development, labor input, irrigation areas and climate classifications to provide a thorough understanding of the snowpack insulation effect for heterogeneous analysis.

Several key findings emerge from our analysis. First, our results demonstrate that FDD has a significantly negative effect on winter wheat yields and its magnitudes increase from lower to higher quantiles of the winter wheat yields distribution. Second, by accounting for the interaction term between FDD and SCF, our results show that FDD*SCF has a significantly positive effect at higher quantiles of the winter wheat yields distribution, indicating that the negative effect of FDD will be offset and the effect of snowpack insulation exists at higher quantiles of the distribution. Third, we show that high GDD in spring has the highest positive effect on winter wheat yields among all GDD terms. Our estimates also suggest that snowfall in autumn and winter has a positive effect on winter wheat yields throughout the entire distribution, showing a supportive role of moisture from snowmelt.[23] Lastly, results from heterogeneity analysis demonstrate that the snowpack insulation effect on winter wheat yields persists at higher quantiles of the winter wheat yields distribution in regions with low economic development, regions with

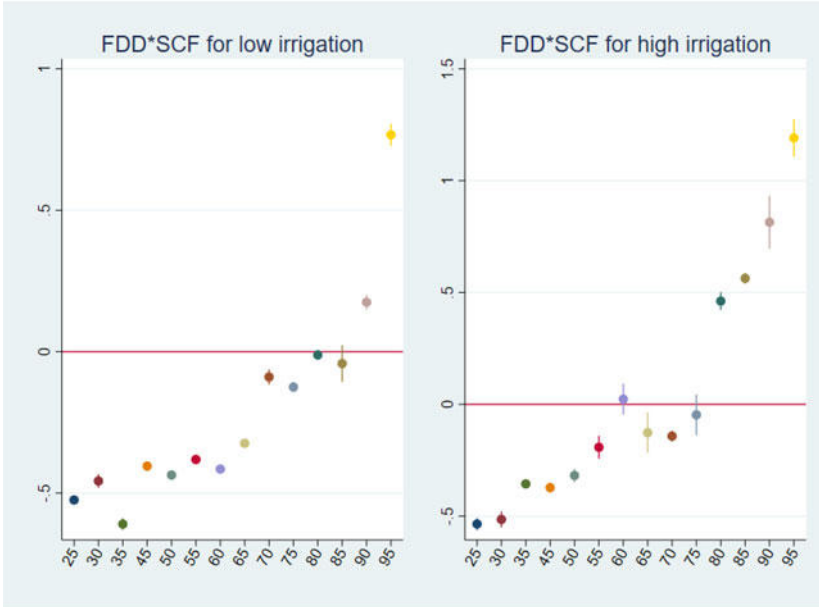
[22] We obtained climate classifications from Rubel and Kottek (2010).

[23] Previous scientific literature has shown that snowmelt water could have a positive impact on crop yields (Biemans *et al.*, 2019; Qin *et al.*, 2020).

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(a) QRPD estimation results: FDD

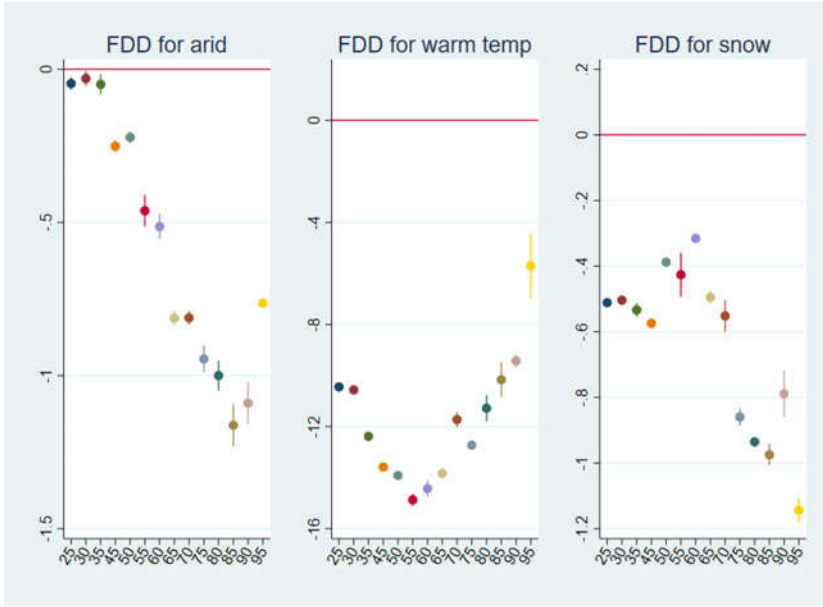


(b) QRPD estimation results: FDD*SCF

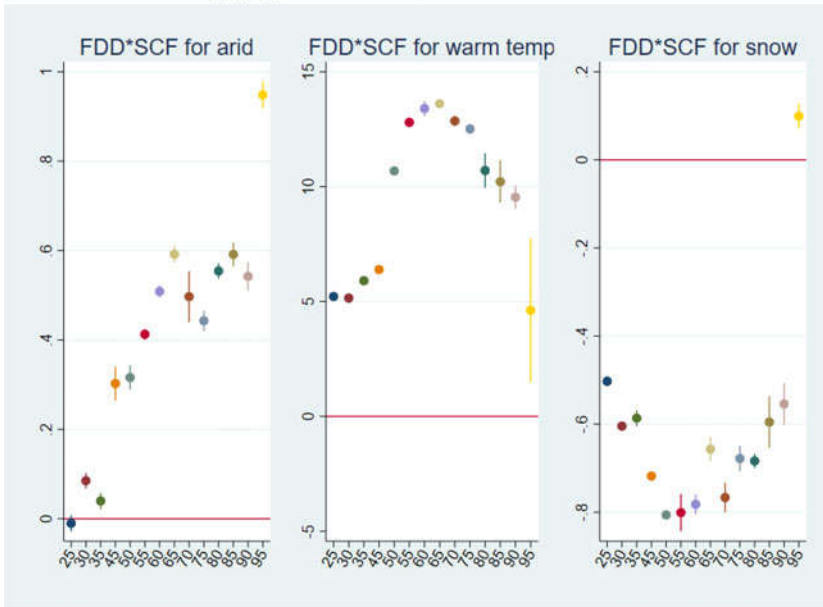
Figure 7. Heterogeneity analysis: irrigation intensity. (a) QRPD estimation results: FDD, (b) QRPD estimation results: FDD*SCF.

Notes: The dependent variable in all regressions is the mean of winter wheat yields. The top two panels show QRPD estimation results of FDD for low and high irrigation intensity. The bottom two panels show QRPD estimation results of FDD*SCF for low and high irrigation intensity.

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(a) QRPD estimation results: FDD



(b) QRPD estimation results: FDD*SCF

Figure 8. Heterogeneity analysis: climate classifications. (a) QRPD estimation results: FDD, (b) QRPD estimation results: FDD*SCF.
Notes: The dependent variable in all regressions is the mean of winter wheat yields. The top three panels show QRPD estimation results of FDD for arid, warm temp and snow climate classifications. The bottom two panels show QRPD estimation results of FDD*SCF for arid, warm temp and snow climate classifications.

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low labor input, regions with low irrigation areas as well as regions with the climate of arid and warm temperature.

This study has important policy implications. First, given the fact that a unit increase in FDD leads to a loss of 2.37 kg/hectare in winter wheat yields, temporary defensive actions could be essential in mitigating the negative effect of freezing stress. For example, farmers could choose to use plastic mulch to protect crops from freezing cold and increase crop yields (Liu and Siddique, 2015; Gao *et al.*, 2019). Second, a back-of-the-envelope calculation suggests that preserving the snowpack insulation effect could bring meaningful economic benefits. For example, our estimate indicates that a unit increase in FDD leads to a loss of 2.37 kg/hectare, which would result in a winter wheat yield loss of 128.93 kg/ha from 2015 to 2016 or a total loss of 2.919 million tons on a national scale.[24] Consequently, the monetary value of the winter wheat yields loss on the national scale for each year would be approximately 1084.85 million dollars.[25] Third, as the snowpack insulation effect exists, especially for higher quantiles of winter wheat yields distribution, temporary practices to increase the SCF could be implemented to increase the snowpack insulation effect against the impact of freezing temperature. For example, local government could implement artificial snow under proper meteorological conditions to increase the SCF to keep a stronger snowpack insulation effect. Lastly, results from heterogeneity analysis indicate that temporary and protective measures could be implemented particularly for specific regions with low economic development, regions with low labor input, regions with low irrigation areas as well as regions with climate of arid and warm temperature.

Supplementary material. The supplementary material for this article can be found at https://doi.org/10.1017/S1355770X25000154

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Competing interests. The authors declare none.

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[24] According to the China Agricultural Statistical Yearbook, China cultivated an average of 22.64 million hectares of winter wheat in 2016. From 2015 to 2016, FDD increased from 339.43°C to 393.83°C, resulting in a winter wheat yield loss of 128.93 kg/ha ((393.83 − 339.43) ∗ 2.37 = 128.93). This translates into a total loss of 2.919 million tons of winter wheat yields in China (128.93 ∗ 22.64/1000 = 2.919).

[25] As the Chinese government's minimum purchase price for wheat produced in 2016 was set at $371.65 per ton, the monetary loss of winter wheat yields on a national scale from 2015 to 2016 due to FDD increases would be 1084.85 million dollars (2.919 ∗ 371.65 = 1084.85).

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